Digital Sign

Students will apply their knowledge of using the light matrix and programming sounds to design, build, and program a digital sign for advertisement.

45-90 min. Beginner Grades 6-8

Teacher Support

Key objectives

Students will:

- Describe the function of hardware and software.
- Program lights and sounds to communicate a message.

Things you will need

- SPIKE Prime sets
- Device with SPIKE App installed
- Student journal
- Sticky notes
- Scissors
- Colored paper

Educational standards

CSTA

2-CS-02
Design projects that combine hardware and software components to collect and exchange data.
2-AP-13
Decompose problems and subproblems into parts to facilitate the design, implementation, and review of programs.
2-AP-16
Incorporate existing code, media, and libraries into original programs, and give attribution.
2-AP-19
Document programs in order to make them easier to follow, test, and debug.

CCSS ELA

SL.8.1
Engage effectively in a range of collaborative discussions (one-on-one, in groups, and teacher-led) with diverse partners on grade 8 topics, texts, and issues, building on others' ideas and expressing their own clearly
SL.8.4
Present claims and findings, emphasizing salient points in a focused, coherent manner with relevant

evidence, sound valid reasoning, and well-chosen details; use appropriate eye contact, adequate volume, and clear pronunciation

RST.6-8.3

Follow precisely a multistep procedure when carrying out experiments, taking measurements, or performing technical tasks

L.8.6

Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression

NGSS

MS-ETS1-2

Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

Vocabulary

Advertise

Question to investigate

How can light and sound be combined to communicate an idea?

Prepare

Check to make sure SPIKE Prime hubs are charged, especially if connecting through Bluetooth.

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1. Engage

Kate and Kyle created a popcorn stand, but no one seems to be buying their popcorn. Kate has an idea to create a sign that will get people's attention. First, they think about making a paper sign. Then, they think about the cool flashing sign they always see at the fair. Can your team help Kate and Kyle create a digital sign for their popcorn stand?

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2. Explore

Students will design and build a sign that uses the light matrix then program their sign to display a message to advertise popcorn or another item. Consider allowing students choice in what item they might be selling and need to advertise.

Challenge students to think of different ways that the light matrix on their hub can be used to create a digital sign. Show images or videos of digital signs. Ask student to name any examples of digital signs they have seen before.

Students should sketch their sign design including writing out the message they play to display on the light matrix. Students need to consider the design of the sign so it will hold the hub in a way that allows their message to be displayed.

> **Hint:** Consider orientation of the hub. Students should consider if they want words and images included in their sign.

Remind students to test their program several times to ensure the message is clear. Students should consider tradeoffs in their design such as the length of a message (easy to read) while making sure their advertisement is clear (people understand the message).

**Requirements for this challenge: **

- The light matrix cannot stand on its own. Students must build a frame for it.
- The digital sign must be advertising popcorn or the item of their choice.
- Prior to writing code, students must show a sketch of their design and write an explanation of what they want to the code.
- Students should use the comment feature in their Python code to explain what the lines of code are meant to do.

  Note: If students struggle with programing their light matrix to create a digital sign remind them to use the Knowledge Base. You can find the information under Hub, then choose Light Matrix and look for show_image() for inspiration on ideas and support in programming.

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3. Explain

Students should share their sign design and explain how it works.

Ask students:

- How did you program your digital sign? Ask students to share their program comments to explain.
- What decisions did you have to make while creating your sign?
- What were areas that you had to debug or troubleshoot?
- What was difficult about this challenge?

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4. Elaborate

Students can also draw attention to the popcorn stand or item of their choice by adding sound to their sign. Ask students to add a melody using the beep sound to their sign that plays only while the lights are used on the hub (i.e. their message is scrolling).

Allow students to share their final design with lights and sounds with other groups.

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5. Evaluate

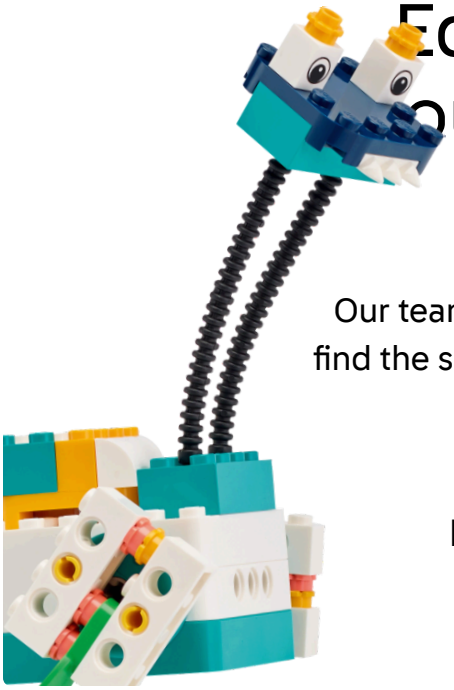
Teacher Observation:
Discuss the program with students. Ask students questions like:

- What libraries were imported for your digital sign program?
- Where did you hear the sounds and the beeps?

Self-Assessment:
Have students answer the following in their journals:

- How can you use beeps and sounds to communicate?

- What characteristics of a good teammate did I display today?
- Ask students to rate themselves on a scale of 1-3, on their time management today.
- Ask student to rate themselves on a scale of 1-3, on their materials (parts) management today.



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